VR-Numbers.

Operational Superstructures of Integers, Rationals, Reals, and Complex Numbers over the Natural Numbers of VR

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2026

Version 1.0.3

Abstract

Building on VR (Reznik, 2026, DOI: 10.5281/zenodo.20212092), a formal axiomatization of arithmetic with three primitives {∅, →, t} and four axioms, we develop *VR-Numbers* — a program of operational superstructures yielding ℤ, ℚ, ℝ, and ℂ over VR's natural numbers ℕ. The construction uses no ordered pairs as primitive objects; each numerical extension is defined as a formal language of expressions with equivalence relations and operations. The entire system rests on the base operation ∅ alone — a nullary operation, not an ontological primitive; all numerical objects — including ℕ — arise as operational constructions of progressively greater depth. The classical actual infinity is replaced by *operational infinity*: A4 (induction) is read as an operational principle, not as a postulate of completed infinite totalities. The two-dimensionality of ℂ is *structurally grounded in* the duality present in axiom A1 of VR (the two generating implications F → F and F → ⊤), rather than postulated through pairs of reals; the algebraic coupling of the two axes requires an additional joining axiom ($i^2 =_{\mathbb{Z}} 1$). ℚ_VR, ℂ_VR, and _VR are shown to be isomorphic to the corresponding standard structures; ℝ_VR is isomorphic to the field of computable real numbers — a countable subfield of classical ℝ — reflecting the operational character of the construction.

Version 1.0.2 (2026) adds Part VIII collecting methodological observations from the Lean 4 formalisation of Parts II–V (companion publication, Zenodo DOI 10.5281/zenodo.20352057). No axioms, definitions, or theorems of Parts I–VII are altered. Version 1.0.3 (2026) reframes the philosophical presentation to the no-ontology position — objects are terms over operations and ∅ is a nullary operation carrying no ontological content; no mathematics is altered.

Keywords: foundations of mathematics, operational extensions, operational infinity, integer construction, rational construction, real numbers, complex numbers, Cauchy sequences, axiomatic duality, operational foundations of numbers, computable analysis.

Part I. Foundation: VR and ℕ

This work builds upon the formal system VR introduced in [Reznik 2026]. We briefly recall its essentials.

Primitives of VR.

Constant: ∅. Binary operator: → (implication). Unary operator: t (succession).

Axioms.

A1 (Generativity). $F \to F = \top$ and $F \to \top = \top$, where F is identified with ∅ at the logical level, and $\top := F \to F$.

A2 (Implication). → is the classical implication on $\{F, \top\}$.

A3 (Succession). $t(x) = x \cup \{x\}$, so $x \in t(x)$ and $x \subset t(x)$.

A4 (Induction). The objects O_n = t^n(∅) exhaust all objects generated from ∅ via t.

Natural Numbers.

$O_0 = \varnothing$, $O_1 = t(\varnothing) = \{\varnothing\}$, $O_2 = t(t(\varnothing)) = \{\varnothing, \{\varnothing\}\}$, …

The natural number n is identified with the expression O_n. Equality is defined via Leibniz's principle: $x = y \iff \forall p: p(x) \leftrightarrow p(y)$. VR has been shown to be arithmetically equivalent to Peano arithmetic, and consistent relative to ZF.

Ontological Status.

In VR there is no ontological primitive: ∅ is itself a nullary operation — the base of the construction — characterised solely by $\forall x\ \neg(x \in \varnothing)$. All other objects, the natural numbers included, are terms over the operations: O_n is not an independent existing thing but the result of n applications of the succession operator t to the base ∅. The mathematical content of n is its operational history.

This operational position is the foundation of the present work. It asserts no ontology — nothing simply "exists", not even ∅, which is a nullary operation — and treats every construction, ∅ included, as operations: objects are terms over the operations, of progressively greater depth.

Part II. Integers ℤ as an Operational Superstructure

We construct integers as formal expressions over ℕ with prescribed equivalence and operations.

II.1. The Formal Language of Subtraction

We extend the language of VR by a binary symbol ⊖. An expression of the form

$$a \ominus b, \text{ where } a, b \in \mathbb{N}$$

is a syntactic object intended to denote the operation "subtract b from a". This is not an ordered pair, but a formal record of an operation.

II.2. Equivalence Relation

Two expressions are declared equivalent when they would yield the same result, using only the addition operation already defined on ℕ:

$$a \ominus b \approx c \ominus d \iff a + d = b + c.$$

This relation is reflexive, symmetric, and transitive (direct verification using the commutativity and associativity of addition in ℕ, established in VR).

II.3. Operations

Addition.

$$(a \ominus b) \oplus (c \ominus d) := (a + c) \ominus (b + d).$$

Multiplication.

$$(a \ominus b) \otimes (c \ominus d) := (a\times c + b\times d) \ominus (a\times d + b\times c).$$

Additive Inverse.

$$\ominus(a \ominus b) := b \ominus a.$$

Subtraction.

$$(a \ominus b) \boxminus (c \ominus d) := (a + d) \ominus (b + c).$$

Remark. Subtraction is definable through addition and additive inverse: $(a \ominus b) \boxminus (c \ominus d) = (a \ominus b) \oplus \ominus(c \ominus d)$. We list it explicitly for convenience.

All operations are well-defined on equivalence classes.

II.4. Canonical Form

Every expression $a \ominus b$ can be canonized uniquely:

If $a \geq b$, the canonical form is $(a - b) \ominus 0$.

If $a < b$, the canonical form is $0 \ominus (b - a)$.

II.5. Embedding ℕ → ℤ_VR

The natural number $n \in \mathbb{N}$ embeds into ℤ_VR as the class of $n \ominus 0$.

II.6. Isomorphism Theorem

Theorem. The set ℤ_VR of equivalence classes of expressions $a \ominus b$ with operations $\oplus$, $\otimes$, $\boxminus$ is a commutative ring with unit, isomorphic to the standard ring of integers ℤ.

Proof sketch. Define $\varphi: \mathbb{Z}\_VR \to \mathbb{Z}$ by $\varphi([a \ominus b]) = a - b$. This map is well-defined on classes, respects operations, and is bijective.

Part III. Rationals ℚ as an Operational Superstructure

Rational numbers are constructed analogously to integers, but over ℤ_VR.

III.1. The Formal Language of Division

We introduce a binary symbol ⊘. An expression of the form

$$a \oslash b, \;\text{ where } a, b \in \mathbb{Z}\_VR \text{ and } b \neq 0\_\mathbb{Z}$$

is a syntactic object denoting the operation "divide a by b". The restriction b ≠ 0_ℤ is syntactic and is effectively decidable via the canonical form in ℤ_VR.

III.2. Equivalence Relation

$$a \oslash b \approx\_\mathbb{Q}\; c \oslash d \quad \otimes \quad a \otimes d \approx b \Longleftarrow c,$$

where ℤ is the multiplication operation in   _VR.

III.3. Operations

Addition.

$$(a \oslash b) \boxplus (c \oslash d) := (a \otimes d \oplus b \otimes c) \oslash (b \otimes d).$$

Multiplication.

$$(a \oslash b) \boxtimes (c \oslash d) := (a \otimes c) \oslash (b \otimes d).$$

Subtraction.

$$(a \oslash b) \boxminus\oslash (c \oslash d) := (a \otimes d \boxminus b \otimes c) \oslash (b \otimes d).$$

Division.

$$(a \oslash b) \oslash\oslash (c \oslash d) := (a \otimes d) \oslash (b \otimes c), \text{ provided } c \neq 0\_\mathbb{Z}.$$

III.4. Canonical Form

Every fraction a ⊘ b can be brought to canonical form by making the denominator positive and reducing to lowest terms.

III.5. Embedding ℤ → ℚ_VR

The integer a ∈ ℤ_VR embeds into ℚ_VR as the class of a ⊘ 1_ℤ, where 1_ℤ = 1 ⊖ 0 is the multiplicative identity in ℤ_VR.

III.6. Isomorphism Theorem

Theorem. $\mathbb{Q}$_VR with operations ⊞, ⊠, ⊟⊘, ⊘⊘ is a field, isomorphic to the standard field of rationals $\mathbb{Q}$.

Part IV. Real Numbers $\mathbb{R}$ via Cauchy Sequences

Real numbers require a qualitatively different construction: while $\mathbb{Z}$ and $\mathbb{Q}$ are syntactic constructions over finite expressions, $\mathbb{R}$ requires infinite processes.

IV.1. Functions as Operational Rules

We extend the language of VR by the notion of function $\mathbb{N} \to \mathbb{Q}$_VR. A function a: $\mathbb{N} \to \mathbb{Q}$_VR is an operational rule: a finite description (algorithm) producing, for each $n \in \mathbb{N}$, a value $a(n) \in \mathbb{Q}$_VR. Functions are not introduced as sets of ordered pairs; they are operational primitives — finite specifications of infinite correspondences.

A consequence of this definition is that each function is fully captured by its finite description. The collection of all such finite descriptions is itself operationally countable: any function in VR-Numbers can be enumerated through its description. This will have implications for the relationship between $\mathbb{R}$_VR and the classical $\mathbb{R}$, discussed below.

Operationally, a function $\mathbb{N} \to \mathbb{Q}$_VR is a partial recursive function in the sense of Church–Turing, or equivalently a Type-2 computable function in the framework of computable analysis (Weihrauch, 2000). The construction $\mathbb{R}$_VR therefore coincides with the standard development of computable real numbers via computable Cauchy sequences (Pour-El & Richards, 1989). The novelty here is not technical but foundational: in VR-Numbers, this is not one approach among several but the only available approach, dictated by the operational position.

IV.2. Fundamental (Cauchy) Sequences

A function a: $\mathbb{N} \to \mathbb{Q}$_VR is fundamental (a Cauchy sequence) if:

$$\forall \varepsilon \in \mathbb{Q}\_VR,\ \varepsilon > 0,\ \exists N \in \mathbb{N} \text{ such that } \forall m, n \geq N\text{: } |a(m) ⊟⊘ a(n)| < \varepsilon.$$

IV.3. Equivalence of Sequences

$$a \approx\_C\ b \iff \forall \varepsilon > 0, \exists N\text{: } \forall n \geq N\text{: } |a(n) ⊟⊘ b(n)| < \varepsilon.$$

IV.4. The Construction $\mathbb{R}$_VR

$\mathbb{R}$_VR is defined as the set of equivalence classes of fundamental sequences with respect to ≈_C.

IV.5. Operations

Operations are defined componentwise on representatives:

$$[a] \oplus\_\mathbb{R}\ [b] := [c], \text{ where } c(n) = a(n) ⊞ b(n);$$

$$[a] \otimes\_\mathbb{R}\ [b] := [c], \text{ where } c(n) = a(n) \boxtimes b(n);$$

$$[a] \boxminus\_\mathbb{R}\ [b] := [c], \text{ where } c(n) = a(n) \boxminus\oslash\ b(n);$$

$$[a] \oslash\_\mathbb{R}\ [b] := [c], \text{ where } c(n) = a(n) \oslash\oslash\ b(n).$$

IV.6. Embedding ℚ → ℝ_VR

Each rational q ∈ ℚ_VR embeds into ℝ_VR as the class of the constant sequence a(n) = q for all n.

IV.7. Isomorphism Theorem

Theorem. ℝ_VR is a real closed ordered field. It is isomorphic to the field of computable real numbers — a countable subfield of the classical real line ℝ — and contains all reals that admit a finite algorithmic description.

Remark. Strictly speaking, ℝ_VR is not isomorphic to the entire classical ℝ. Classical ℝ has cardinality of the continuum; ℝ_VR, consisting of operational rules each given by a finite description, is countable. The relationship is therefore: ℝ_VR is isomorphic to the computable reals, which constitute a countable real closed subfield of classical ℝ. All real numbers encountered in practice — algebraic numbers, rationals, π, e, and indeed any real definable by a finite description — are captured by ℝ_VR. Non-computable reals of classical ℝ have no place in the operational setting of VR-Numbers, as they correspond to no finite algorithmic description. This aligns VR-Numbers with the constructive tradition (Bishop, 1967; Martin-Löf, 1984) and remains consistent with the philosophical position that mathematical objects are operations, not transcendent entities.

IV.8. Operational Character of ℝ

A real number in VR is, formally, an algorithm — a finite description specifying how to compute arbitrarily close rational approximations. The square root of 2, for instance, is represented by an algorithm (such as Newton's method) that, given any precision, produces a rational approximation within that precision. This view aligns naturally with constructive mathematics.

Part V. Complex Numbers ℂ via the Duality of A1

The construction of complex numbers introduces the central new conceptual element of this work. In standard mathematics, ℂ is built as ordered pairs of real numbers (a, b) with a special multiplication rule, and the two-dimensionality of ℂ is postulated through the choice of pairs. We propose instead that the two-dimensionality of ℂ is structurally grounded in the logical foundation of VR — specifically, in the duality present in axiom A1.

V.1. The Duality of A1

Axiom A1 of VR states:

$$F \to F = \top,$$

$$F \to \top = \top.$$

These are two distinct generating facts: F generates both values via implication. The two implications represent two distinct movements from F:

$F \to F$: a movement returning to F itself (self-reference at the logical level).

$F \to \top$: a movement to the opposite value $\top$.

We propose to interpret these two implications as two independent axes of numerical extension.

V.2. The Two Axes

The axis $F \to F$ is the real axis: along this axis, the constructions $\mathbb{N} \to \mathbb{Z} \to \mathbb{Q} \to \mathbb{R}$ developed above unfold.

The axis $F \to \top$ is the imaginary axis: along this axis, new operational expressions of the form $b \cdot i$ live, where i is an operational marker indicating the imaginary axis.

The marker i is not a number nor an object — it is a syntactic indicator of which axis a value belongs to.

V.3. The Joining Rule

To make $\mathbb{C}$ a single algebraic structure rather than two disjoint copies of $\mathbb{R}$, the two axes must be coupled. This coupling is captured by the rule:

$$i \otimes i := \ominus 1.$$

This rule is postulated as an axiom of joining: applying the imaginary marker twice is equivalent to negation along the real axis. Without this rule, the two axes would remain independent; with it, they are intertwined into the algebraic structure of $\mathbb{C}$.

V.4. Complex Numbers as Operational Expressions

A complex number is a formal expression

$$a \oplus b \cdot i, \text{ where } a, b \in \mathbb{R}\_VR.$$

This is not an ordered pair (a, b). It is a syntactic record of two values along two distinct axes.

V.5. Equivalence

$$a \oplus b \cdot i \approx c \oplus d \cdot i \iff a = c \text{ (in } \mathbb{R}\_VR\text{) and } b = d \text{ (in } \mathbb{R}\_VR\text{)}.$$

V.6. Operations

Addition.

$$(a \oplus b \cdot i) \oplus\_\mathbb{C} (c \oplus d \cdot i) := (a \oplus c) \oplus (b \oplus d) \cdot i.$$

Multiplication (using $i \otimes i = \ominus 1$).

$$(a \oplus b\cdot i) \otimes\_\mathbb{C}\ (c \oplus d\cdot i) := (a\otimes c \boxminus b\otimes d) \oplus (a\otimes d \oplus b\otimes c)\cdot i.$$

Conjugation.

$$\mathrm{conj}(a \oplus b\cdot i) := a \oplus (\ominus b)\cdot i.$$

Modulus.

$$|a \oplus b\cdot i| := \sqrt{(a\otimes a \oplus b\otimes b)}.$$

V.7. Embedding ℝ → ℂ_VR

Each real $a \in \mathbb{R}$_VR embeds into ℂ_VR as the expression $a \oplus 0\cdot i$.

V.8. Isomorphism Theorem

Theorem. ℂ_VR with the operations defined above is a field, isomorphic to the standard field of complex numbers ℂ (with ℝ replaced by ℝ_VR; the resulting field is isomorphic to the field of computable complex numbers).

V.9. Discussion: Foundation of Two-Dimensionality

In the standard treatment, the dimensionality of ℂ is a postulate: ℂ is defined as $\mathbb{R} \times \mathbb{R}$ with a specific multiplication, and there is no deeper justification for why ℂ has exactly two dimensions.

In VR-Numbers, the two-dimensionality of ℂ is structurally grounded in the logical foundation of the system. Axiom A1 contains exactly two generating implications, $F \to F$ and $F \to \top$. These two implications correspond to two distinct axes of numerical construction. The dimensionality of ℂ is therefore not arbitrary; it reflects the dual generative structure already present in the logical layer of VR.

Scope of this claim. The duality of A1 provides two distinct generative directions but does not by itself entail their algebraic coupling. The rule $i^2 = \ominus 1$ is an additional postulate joining the two axes into a single field. Thus A1 *motivates* the two-dimensionality of ℂ; the field structure on these two dimensions requires the joining axiom. The conceptual content of the claim is that the two-dimensionality is not arbitrary — it has a logical origin — not that the entire algebraic structure of ℂ is derivable from A1 alone.

Position relative to higher-dimensional algebras. The duality of A1 yields exactly two generative directions at the logical level. The theorems of Frobenius (1878) and Hurwitz (1898) show that further normed division algebras over ℝ — quaternions ℍ and octonions 𝕆 — require successive weakening of algebraic properties (commutativity, then associativity). This is consistent with the position that A1 as a logical primitive does not further subdivide: extensions beyond ℂ require leaving the purely logical layer and accepting algebraic compromises. Whether such higher-dimensional algebras admit analogous derivations from deeper logical structure remains an open question (see Part VII).

This conceptual claim — that the structure of complex numbers is not arbitrary but reflects how implication operates on falsity — is the principal philosophical contribution of the present work. The two generating movements from F (one returning to F itself, one moving to ⊤) are precisely the two axes that, when joined by the rule $i^2 = \ominus 1$, yield the field of complex numbers.

Part VI. The Ontological Position of VR-Numbers

VR-Numbers ascribes no ontology: there is no ontological primitive — ∅ is a nullary operation, characterised solely by ∀x ¬(x ∈ ∅). All objects — natural, integer, rational, real, and complex numbers — are operational constructions (terms over the operations) of progressively greater depth.

VI.1. No Ontological Primitive: the Base as a Nullary Operation

∅ is not an ontological primitive: it is the nullary base operation of the system and carries no ontological content — its entire characterisation is ∀x ¬(x ∈ ∅), a fact about the relation ∈, not a description of any contents. Nothing is "prior to operations": ∅ is itself an operation. Every object of VR or VR-Numbers is a term composed of operations.

The succession operator t is not itself an object but a rule of operation. The natural numbers O_n = t^n(∅) are records of successive applications of t. Their content is fully captured by their operational history: O_3 is "three applications of t to ∅", and this is the totality of what O_3 is.

VI.2. Depths of Operationality

All numerical systems constructed in this work are operational, but they exhibit different depths of operational character:

System	Operational Character
∅	The single ontological primitive — that which is, prior to all operation
ℕ	Direct applications of t to ∅: O_n = t^n(∅)
ℤ	Finite syntactic expressions a ⊖ b over ℕ, on the F → F axis
ℚ	Finite syntactic expressions a ⊘ b over ℤ, on the F → F axis
ℝ	Algorithms producing infinite sequences of rationals, on the F → F axis
ℂ	Two-axis syntax over F → F and F → ⊤, joined by $i^2 = \ominus 1$

VI.3. Operational Infinity

The axiom A4 of VR — the principle of induction — is conventionally read as postulating actual infinity: the completed totality of all natural numbers existing as an object. We reject this reading. Within the operational position of VR-Numbers, A4 does not assert the existence of any infinite object. It formulates an operational principle: any property holding for $\varnothing$ and preserved under application of t holds for every result of iterated application of t. The quantifier "for all n" in A4 is operational, not existential — it expresses universal applicability of an operation, not membership in a completed set.

We call this *operational infinity*, in contrast to the classical notion of actual infinity. Operational infinity is mathematically equivalent to actual infinity: every theorem provable using actual infinity in classical mathematics is provable using operational infinity in VR-Numbers. The two notions agree on consequences. They differ in what they assert to exist: actual infinity asserts that an infinite totality exists as an object; operational infinity asserts only that an infinite operation is available.

This distinction is significant. The chief philosophical objection to actual infinity, raised since Aristotle and reformulated by intuitionists and constructivists, is the rejection of completed infinities as objects. Operational infinity bypasses this objection: it does not postulate that the totality of natural numbers is a thing. It postulates that the procedure of unbounded counting from $\varnothing$ is available to us, and that any property preserved by this procedure applies universally. The resulting mathematics is the same; nothing is posited as a completed object.

Within VR-Numbers, all references to "infinite" objects — the natural numbers $\mathbb{N}$ as a domain, the integers $\mathbb{Z}$, the rationals $\mathbb{Q}$, the reals $\mathbb{R}$ as Cauchy sequences, the complex numbers $\mathbb{C}$ — are to be understood operationally. None of these collections is an object. Each is a domain of applicability for operations defined over $\varnothing$.

Accordingly, the quantifier notations used throughout this work — expressions of the form "$a, b \in \mathbb{N}$" or "$\forall\varepsilon \in \mathbb{Q}$_VR, $\varepsilon > 0$" — are operational. They mean: "for any concrete expression of the indicated form". They do not assert the existence of $\mathbb{N}$ or $\mathbb{Q}$_VR as completed objects. The set-theoretic notation is retained for convenience and conformity with standard mathematical practice; the intended content is operational throughout.

Note on terminology. Throughout this work, set-theoretic vocabulary ("equivalence class", "set", "field", "subfield") is retained for conformity with standard mathematical practice. Within the operational setting of VR-Numbers, every such phrase is to be read operationally: an "equivalence class" is the operational rule for recognizing equivalence, not a completed object containing its members; a "field" is the closed system of operations and their results, not a completed collection of values. The forthcoming work *VR-Sets* will provide the operational reconstruction of set-theoretic vocabulary itself. Until then, the standard vocabulary is used with the operational reading understood.

VI.4. The Gradient of Operationality

The structure of VR-Numbers is a gradient: each successive numerical system corresponds to a deeper operational construction over $\varnothing$.

Natural numbers are the most direct operational result: a single operator t is applied repeatedly to the primitive $\varnothing$.

Integers and rationals introduce a layer of formal syntax: expressions over natural numbers, with equivalence relations defining when distinct expressions denote the same operational result.

Real numbers add infinite operational depth: each real number is itself a finite description (algorithm), but the operation it specifies is infinite — producing arbitrarily close rational approximations.

Complex numbers add a second axis: the duality of A1 generates two independent operational directions, and $\mathbb{C}$ is the syntactic structure on these two axes joined by the multiplication rule.

This gradient reflects the historical development of mathematics. Natural numbers have been recognized in all cultures since antiquity, while negative, rational, real, and complex numbers were progressively introduced as the operational needs of mathematics demanded them. The conceptual difficulty historically accompanying each extension — *numeri ficti* for negatives, the irrationality crisis in Greek mathematics, the long dispute over imaginary numbers — reflects, in retrospect, the increasing operational depth of these constructions.

VI.5. What Does Not Exist as an Object

In VR-Numbers, the following are not introduced as primitive objects:

> Ordered pairs. Pairs appear nowhere in the construction as independent objects; pairing is replaced by syntactic juxtaposition ($a \ominus b$, $a \oslash b$, $a \oplus b \cdot i$).
>
> Negative numbers. The integer -3 is a notation for the operational class $\{n \ominus (n+3) : n \in \mathbb{N}\}$, not an independent thing.
>
> Fractions. The rational 1/2 is a notation for the class of equivalent division expressions, not a separate object.
>
> Real numbers. A real number is an equivalence class of Cauchy sequences, which are themselves operational rules — finite specifications of infinite processes — not sets of ordered pairs.
>
> Complex numbers. A complex number is a syntactic expression over two operational axes, not a two-dimensional point in a plane.

This minimalism is the philosophical position of VR-Numbers: nothing is — all is doing.

VI.6. Consistency Relative to ZF

Each construction ($\mathbb{Z}$_VR, $\mathbb{Q}$_VR, $\mathbb{C}$_VR) is isomorphic to its standard counterpart. The field of computable real numbers is a definable subfield of $\mathbb{R}$ in ZF (Pour-El & Richards, 1989), and $\mathbb{R}$_VR is isomorphic to it; therefore $\mathbb{R}$_VR has a ZF-model. Combined with the standard ZF-models for $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{C}$ and the consistency of VR relative to ZF [Reznik 2026], the full system of VR-Numbers is consistent relative to ZF.

Part VII. Open Questions

(1) Formalization in Type Theory.

The constructions of VR-Numbers — particularly the operational view of functions in the definition of $\mathbb{R}$, and the operational marker i in $\mathbb{C}$ — are naturally suited to constructive type theories such as Martin-Löf's. A formalization in Coq, Agda, or Lean would provide machine verification and explicit categorical semantics.

(2) Algebraic Closure Beyond $\mathbb{C}$.

By the Fundamental Theorem of Algebra, $\mathbb{C}$ is algebraically closed. The theorems of Frobenius (1878) and Hurwitz (1898) further show that $\mathbb{C}$ is the largest finite-dimensional commutative associative normed division algebra over $\mathbb{R}$ — quaternions $\mathbb{H}$ and octonions $\mathbb{O}$ require successive weakening of algebraic properties. Whether these higher-dimensional algebras admit analogous derivations from deeper logical structure — and whether such structure could be located in VR-style logical primitives — remains an open question.

(3) The VR-Sets Program: Operational Set Theory.

A natural extension of the operational position is an operational reconstruction of set theory itself, provisionally named *VR-Sets*. The central idea is that the membership relation $x \in y$ admits an operational reading as "x was used in the construction of y", distinguishing *reference* from *physical containment*. Under this reading, the expression $x \in x$ is not paradoxical: it expresses that x is a result of an operation referring to its own name, analogous to recursive structures in programming. This opens a possible unification of well-founded (ZFC-mode) and non-well-founded (ZFA-mode) set theories as two operational modes of a single system, with bisimulation (Aczel, 1988) as the natural equivalence criterion in the ZFA-mode. The technical development of VR-Sets — including the operational reformulation of the ZF axioms, the status of the axiom of choice in operational vs. non-operational regimes, and the relationship between A1's duality and Tarski's theorem on the undefinability of truth — is the subject of forthcoming work.

(4) Computational Realization.

Each level of VR-Numbers admits a direct computational realization, making the system suitable as a foundation for exact-arithmetic computation systems.

Part VIII. Notes from the Lean 4 Formalisation

The constructions of Parts II–V have been formalised in Lean 4 (Reznik, 2026, DOI 10.5281/zenodo.20352057), building on the prior formalisation of Part I (Reznik, 2026, DOI 10.5281/zenodo.20324240). The formalisation comprises approximately 3,600 lines of Lean code across four files (Integers.lean, Rationals.lean, Reals.lean, Complex.lean), with no use of *sorry* or *admit*. All axiom dependencies are documented. The formalisation surfaced several observations that refine or extend the preprint's account. The present Part collects eight of these, organised into three groups: hidden lemmas (observations the preprint absorbs into phrases like "direct verification"), structural boundaries (places where the formalisation makes visible a transition that the preprint addresses qualitatively), and methodological patterns (recurring tactical choices that maintain the axiom-minimality of the formalisation). These observations do not alter any axiom, definition, or theorem of Parts I–VII; they document the relation between the preprint and its Lean realisation.

VIII.1. Hidden lemma in §II.2: cancellation of addition

Section II.2 establishes reflexivity, symmetry, and transitivity of the equivalence $a \ominus b \approx c \ominus d$ in a single phrase: "direct verification using the commutativity and associativity of addition in $\mathbb{N}$." In the Lean formalisation this phrase splits into three steps of unequal structural cost. Reflexivity reduces to one application of T1 (commutativity). Symmetry requires two applications of T1 in a chain. Transitivity, however, requires a separate lemma not stated in T1–T4: the *right cancellation law* for addition on VR-naturals: $a + c = b + c \to a = b$. This is not a consequence of T1–T4 by rewriting; it is proved by induction on c using the injectivity of the successor (the Peano principle P4 of Part I). The preprint absorbs this cancellation into the phrase "direct verification"; the formalisation factors it out as an explicit lemma whose proof retraces the inductive structure of $\mathbb{N}$.

VIII.2. Hidden lemma in §II.3: left distributivity of multiplication

Theorem T3 of Part I states right distributivity: $a \times (b + c) = a{\times}b + a{\times}c$. The symmetric left distributivity $(a + b) \times c = a{\times}c + b{\times}c$ is not in T1–T4 and is not derivable by T3 alone, since commutativity of multiplication is not among T1–T4 either. In informal mathematics the two forms are interchanged silently. In the Lean formalisation, the proof of well-definedness of multiplication on integers (§II.3) requires left distributivity explicitly; we prove it by induction on the right argument using T3 and a combinatorial swap lemma. The observation is twofold: (i) the preprint's T1–T4 do not commit to commutativity of multiplication on $\mathbb{N}$, and (ii) the formalisation makes this restraint visible by requiring an additional lemma at the first place where it matters.

VIII.3. Hidden lemma in §III.2: multiplicative cancellation as a nonzero condition

Transitivity of the equivalence $a \oslash b \approx c \oslash d$ (§III.2) relies on the cancellation law $xz = yz \to x = y$ when $z \neq 0$ — the integral-domain property of $\mathbb{Z}$. The preprint states the transitivity as "direct verification using multiplication in $\mathbb{Z}$_VR." In Lean this expands into three technical components: (i) propagation of the nonzero condition through the isomorphism $\mathbb{Z}$_VR $\cong$ $\mathbb{Z}$; (ii) a cancellation lemma for $\mathbb{Z}$ itself, which we prove constructively via the absolute value Int.natAbs without invoking mathlib's general IntegralDomain infrastructure (which would pull in Classical.choice unnecessarily); and (iii) a factored polynomial identity, proved by ring, in place of the linear_combination tactic. The observation: integral-domain cancellation on $\mathbb{Z}$ is constructive in itself; the dependence on Classical.choice appears only through the infrastructural generality of mathlib, not through the cancellation property at this concrete type.

VIII.4. Structural boundary: Classical.choice between ℤ and ℚ

The formalisation of ℤ_VR (Parts II of the preprint) is fully constructive: every theorem depends only on the standard quotient axiom Quot.sound (and propositional extensionality propext). At ℚ_VR the situation changes: every theorem mentioning addition or multiplication on rationals depends on Classical.choice. The reason is structural to Lean 4 Core: the Rat type carries a field reduced : num.natAbs.Coprime den, so every Rat.add and Rat.mul normalises its result through a coprimality proof that depends on Classical.choice via Nat.gcd. This dependency is not introduced by the formalisation; it cannot be removed by lemma substitution or tactic choice. The classical-versus-constructive boundary in this formalisation therefore falls between ℤ_VR and ℚ_VR. This is a structural feature of Lean 4 Core's representation of rationals, not a property of the rationals themselves: an alternative Rat without normalisation (a plain quotient of pairs) would remain constructive. The boundary corresponds precisely to the transition from operations performed *syntactically* (a + b on ℤ) to operations requiring a *canonical representative* (gcd-reduction of a/b on ℚ) — the same transition that §VI.4 describes informally as a deepening of operationality.

VIII.5. Structural boundary: the quotient construction becomes vacuous at ℂ

Each of ℤ_VR, ℚ_VR, and ℝ_VR is built by the same three-step pattern: a syntactic record, a nontrivial equivalence relation, and a quotient. The equivalence collapses syntactically distinct representatives into one mathematical object (cross-addition, cross-multiplication, Cauchy ε-convergence). At ℂ_VR the equivalence degenerates: a ⊕b·i ≈ c ⊕d·i iff a = c and b = d (§V.5) — componentwise equality, which in the formalisation coincides with Lean's built-in equality on a two-field structure. The quotient becomes vacuous, and ℂ_VR is the first type in the cycle realised without Quotient. The methodological cost of the quotient construction, distributed silently across ℤ, ℚ, ℝ, becomes visible at ℂ by its absence. Substantively: the depth of operationality at ℂ_VR (§VI.4) is carried not by the equivalence relation but by the two-dimensionality grounded in A1 and the joining rule i^2 = ⊖1 (§V.3, §V.9).

VIII.6. Structural boundary: inexpressibility of computability on ℝ_VR

Section IV.1 stipulates that functions ℕ → ℚ_VR are operational rules — finite descriptions producing the values. Section IV.7 then concludes that ℝ_VR is isomorphic to the field of computable real numbers, a countable subfield of classical ℝ. The Lean type Nat → ℚ_VR, however, contains every total Lean function of that signature, including those whose existence requires Classical.choice; this type has the cardinality of the continuum. The formalisation therefore proves an isomorphism between ℝ_VR and the full classical ℝ, not the computable subfield. Lean 4 has no type-level predicate distinguishing computable from non-computable functions; the operational restriction of §IV.1 is metatheoretic relative to Lean. Within the present cycle (VR-Numbers) the recommended interpretation is: the formalisation reflects faithfully the *construction* of ℝ_VR via Cauchy sequences (§IV.1–§IV.6), and additionally proves a structural isomorphism with classical ℝ; the *restriction* to computable reals (§IV.7) is preserved as metatheoretic commentary. This is a concrete instance where VR's operational restriction is strictly stronger than Lean 4 can express in a single type construction. The forthcoming *VR-Forms* provides a formal register in which the relation between operational and non-operational objects can be addressed directly.

VIII.7. Methodological pattern: tactics cleaner than named lemmas

Several lemmas in mathlib carry latent Classical.choice dependencies through algebraic typeclass infrastructure: mul_zero, zero_mul, mul_right_cancel$_0$, linarith on integers, and the linear_combination tactic itself all introduce Classical.choice when used on concrete numeric types. The same statements can be discharged on those concrete types by tactics that act directly on the type (ring, omega, manual factored identities) without invoking the general infrastructure, and remain Classical-free. The pattern recurred at least four times during the formalisation of Parts II and III. The lesson is counter-intuitive: a tactic appears to be less explicit than a named lemma, but on concrete numeric types it is often more axiom-minimal, because it bypasses the typeclass generality where Classical.choice resides. For axiom-minimal formalisation on concrete number systems, prefer tactics to named algebraic lemmas; the symbolic gain of an explicit lemma may come at the cost of an additional axiom in the dependency closure.

VIII.8. Methodological pattern: change vs simp for definitional equality

At three independent points in the formalisation of $\mathbb{R}$_VR (Part IV), automatic simplification (simp) drove the goal into a form syntactically incompatible with the intended rewrite: the strict order on Rat unfolded through a boolean comparison (Rat.blt) breaking the typeclass for less-than; abs_neg under simp triggered a ring-normalisation that hid the rewrite target; Real.mk_eq under simp reshaped the ε-distance into a form using a different CauSeq operator. In each case the resolution was the same: replace simp-driven unfolding with the change tactic, which restates the goal in a definitionally equal but syntactically intended form without performing any reduction. The pattern is related to VIII.7: in both cases, less automation (a weaker tactic) yields a more stable proof than more automation (simp, named lemma) precisely because it preserves syntactic control over the goal. For a formalisation aiming at axiom-minimality and proof stability, the lighter tactic is preferable where it suffices.

Summary

VR-Numbers extends the formal system VR with operational superstructures yielding $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$, and $\mathbb{C}$. The work advances three principal theses:

First, no ontology: there is no ontological primitive — ∅ is a nullary operation; all objects, the natural numbers included, are terms over the operations, of progressively greater depth. No ordered pairs are introduced as primitive objects.

Second, actual infinity is replaced by operational infinity: the axiom of induction A4 is read as an operational principle expressing universal applicability of an operation, not as a postulate of a completed infinite totality. The two notions agree on mathematical consequences but differ in what they posit as existing.

Third, the two-dimensionality of $\mathbb{C}$ is structurally grounded in the duality of axiom A1: the two generating implications $F \to F$ and $F \to \top$ correspond to the real and imaginary axes, respectively. The algebraic coupling of the two axes through $i^2 = \ominus 1$ is an additional joining axiom; A1 motivates the two-dimensionality, while the field structure requires the joining rule. This provides a structural motivation for a feature standardly postulated through pairs without deeper justification.

The structures $\mathbb{Z}$_VR, $\mathbb{Q}$_VR, $\mathbb{C}$_VR are isomorphic to their standard counterparts; $\mathbb{R}$_VR is isomorphic to the field of computable real numbers (a countable subfield of classical $\mathbb{R}$), reflecting the constructive position of the work. The consistency of the system relative to ZF follows from the consistency of VR established in [Reznik 2026] and from the standard ZF-definability of the computable real numbers (Pour-El & Richards, 1989). The philosophical position is minimal: nothing is — all is doing.

Acknowledgement of AI assistance

The exposition of this preprint and its companion Lean 4 formalisation (Parts II–V) were prepared with the assistance of Claude (Anthropic), in the parent–child review architecture (Variant A) used across the VR Cycle, for drafting, English translation, language polishing, consistency checking, and the Lean development. All mathematical content, definitions, theorems, and philosophical positions are due to the author. Version 1.0.3 removes any ontology of the empty set: $\varnothing$ is taken as a nullary operation, characterised solely by $\forall x\, \neg(x \in \varnothing)$, and every object is a term over the operations — there are only operations. This revision was carried out with Claude Opus 4.8.

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